Fairness and Proxy Bias in Occupation Prediction: Scaling, Masking, and Mechanistic Attribution across Model Families

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Abstract

We investigate gender bias and proxy feature encoding in occupation prediction models trained on the Bias-in-Bios dataset. Our study spans two dimensions: *model capacity*, comparing Pythia generative models (160M–1.4B parameters) across zero-shot, few-shot, and fine-tuned regimes, and *model architecture*, contrasting encoder-based classifiers (DistilBERT, RoBERTa) against generative models of comparable performance. We evaluate both task performance (Macro-F1) and gender fairness (Equalized Odds gap, Demographic Parity) under masked and unmasked conditions, and diagnose bias mechanisms through Integrated Gradients attribution validated by top-K token erasure. Our main findings: (1) fine-tuned models substantially outperform zero/few-shot baselines in both performance and fairness; (2) larger models are not inherently fairer—zero-shot bias *increases* with scale; (3) gender-token masking reduces bias for some professions but produces backfire effects for others, suggesting that implicit proxy signals survive removal of explicit gender tokens; (4) explicit gender tokens are not the primary drivers of bias—they appear with weights 61–289$\times$ smaller than dominant profession tokens, while indirect contextual signals carry the latent gender signal; and (5) proxy bias is model-dependent, the same profession can be stable under one model and unstable under another, though the precise cause (architecture, pretraining, or scale) cannot be isolated from this comparison alone.

1 Introduction

Occupation prediction from biographical text exposes persistent gender stereotypes: models trained on real-world career data reliably associate professions such as *nurse* or *dietitian* with female-typical language patterns, and *surgeon* or *software engineer* with male-typical ones. De-Arteaga et al. [2019] demonstrated on the Bias-in-Bios dataset that scrubbing explicit gender signals does not eliminate these disparities—implicit proxy features survive.

This paper addresses three open questions: how does model *capacity* interact with fairness; how does model *architecture* shape which proxy features are learned; and what are those features precisely.

Contributions. (1) A scaling study across Pythia (160M–1.4B) in zero-shot, few-shot, and fine-tuned regimes; (2) a cross-model comparison of encoder baselines (DistilBERT, RoBERTa) against the best Pythia configuration; (3) an Integrated Gradients proxy audit by profession and gender group, validated via top-K token erasure faithfulness testing; (4) a paired masked/unmasked evaluation measuring the effect of gender-token scrubbing on both task performance and fairness metrics; (5) a cross-model proxy stability analysis comparing attribution patterns across encoder models. To our knowledge, no prior work on the Bias-in-Bios occupation prediction task provides token-level attribution analysis stratified by both profession and gender, nor compares proxy stability across encoder models on this dataset.

Key Findings. Our main findings are summarised in the abstract. Critically, zero-shot bias *increases* with Pythia scale (EO gap $\sim$0.09 at 160M vs. $\sim$0.21 at 1.4B), and explicit gender tokens carry attribution weights 61–289$\times$ smaller than dominant profession tokens—indicating that implicit proxy signals, not surface gender words, are the primary drivers of measured bias. We do not claim these findings generalise beyond the Bias-in-Bios setting.

2 Related Work

Bias in occupation prediction. De-Arteaga et al. [2019] introduced Bias-in-Bios and showed that classifiers exhibit gender disparities even after pronoun scrubbing. Romanov et al. [2019] analysed limits of de-biasing through scrubbing. We extend these findings by identifying and quantifying specific proxy tokens through gradient-based attribution.

Fairness metrics. Hardt et al. [2016] formalised Equalized Odds; Dwork et al. [2012] introduced demographic parity. We evaluate both using a one-vs-rest multiclass extension macro-averaged over occupations.

Attribution methods. Sundararajan et al. [2017] introduced Integrated Gradients satisfying completeness and sensitivity axioms. Attribution methods have been applied to gender bias in coreference [Attanasio et al., 2022] but their systematic use in occupation classification is limited. Critically, no prior work on the Bias-in-Bios occupation prediction task provides token-level proxy word tables with quantified attribution weights, nor examines how those weights shift under gender masking.

Model scaling and fairness. Bommasani et al. [2021] discusses risks of foundation models including bias amplification. Our scaling study provides empirical evidence that zero-shot bias increases with model size in the Pythia family.

Gender masking. Lu et al. [2020] study counterfactual data augmentation for gender bias. Unlike prior work, we measure not only the effect of masking on fairness metrics but also on token-level attribution patterns — providing the first internal view of how occupation prediction models redistribute feature importance when gender signals are removed.

3 Dataset and Preprocessing

3.1 Bias-in-Bios

We use the Bias-in-Bios dataset [De-Arteaga et al., 2019], accessed via HuggingFace Datasets (`LabHC/bias_in_bios`). Bias-in-Bios has become a standard benchmark in the NLP fairness community for studying occupational gender stereotypes [De-Arteaga et al., 2019, Romanov et al., 2019]. Each example contains a biography (`hard_text`), an integer profession label, and a binary gender attribute ($0 \to$ M, $1 \to$ F, provided by the dataset and inferred from pronoun usage in the biography text by the original authors). We restrict to the **top 20 most frequent occupations** computed from the training split, filtering out all other examples. The task is to predict a biography's occupation label—one of these 20 classes—from its free-text content alone; gender is used only as a post-hoc evaluation attribute and never as a model input.

Split	Total	Female	Male
Train	248,141	46.16%	53.84%
Dev	38,198	46.16%	53.84%
Test	95,468	46.17%	53.83%
Total	**381,807**	46.16%	53.84%

Table 1: Dataset split sizes and gender distribution. The near-identical F% across splits (range: 0.01%, i.e. approximately 46.2% female in each) indicates the original dataset was stratified by gender when partitioned.

Table 2 shows the per-profession counts and gender distribution in the training split. Several occupations exhibit strong gender skew: *dietitian* (92.9% F), *nurse* (90.8% F), and *model* (82.7% F) are heavily female-dominated, while *surgeon* (14.8% F), *software engineer* (15.8% F), and *composer* (16.4% F) are heavily male-dominated.

Profession	N (train)	F%
professor	76,748	45.1
physician	26,648	49.4
attorney	21,169	38.3
photographer	15,773	35.7
journalist	12,960	49.5
nurse	12,316	90.8
psychologist	11,945	62.1
teacher	10,531	60.2
dentist	9,479	35.3
surgeon	8,829	14.8
architect	6,568	23.7
painter	5,025	45.7
model	4,867	82.7
filmmaker	4,545	32.9
poet	4,558	49.0
software engineer	4,492	15.8
composer	3,637	16.4
accountant	3,660	36.7
dietitian	2,567	92.9
comedian	1,824	21.1

Table 2: Per-profession training counts and F% (M% = 100 − F%). Distribution is consistent across all splits.

3.2 Evaluation Subset

All models are evaluated on a canonical **3,000-sample test subset**, stratified by profession proportionally to each profession's frequency in the full test set, and sampled from the 95,468-example test split. A shared canonical ID list ensures comparability across all models and regimes.

The 3,000-sample subset was chosen to balance statistical representativeness against two computational constraints: (1) the latency of zero-shot and few-shot inference over the full 95K test split, and (2) the cost of running Integrated Gradients attribution across four conditions (two encoder models × two masking conditions), each requiring a separate forward pass per example. Per-profession sample sizes range from 932 (professor) to 27 (comedian); results for low-frequency professions should be interpreted with caution given the small sample sizes.

3.3 Gender Masking

Two conditions: **Unmasked** (original text) and **Masked** (explicit gender tokens replaced with `[MASK]` for DistilBERT and `<mask>` for RoBERTa, matching each model's native mask token), applied via exact case-insensitive word-boundary regex matching (`\b...\b`). Three categories of tokens are masked: (1) **Pronouns and possessives**: *he, she, his, her, him, hers, himself, herself*; (2) **Titles and honorifics**: *mr, mr., mister, mrs, mrs., miss, ms, ms., madam, madame, ma'am, sir, lady, lord, dame*; (3) **Gendered nouns** (35 terms including *father, mother, husband, wife, son, daughter, brother, sister, actor, actress, businessman, businesswoman, waiter, waitress*, and others). Masking is disabled by default and enabled per experiment. Name masking was not applied for two reasons: (1) any general-purpose name-masking rule introduces uncontrolled noise through false positives and false negatives; and (2) names carry biographical content beyond gender—masking them would conflate the removal of gender signals with a broader loss of biographical information, undermining the internal validity of the masked/unmasked comparison.

4 Methods

4.1 Pythia: Zero-shot and Few-shot

We evaluate Pythia [Biderman et al., 2023] at 160M, 410M, and 1.4B parameters under two inference regimes: zero-shot and few-shot, with and without gender masking applied to the input biography.

Zero-shot: biography + candidate list, no examples. **Few-shot**: $K = 4$ exemplars pairs prepended to prompt.

Since Pythia is a causal language model rather than a classifier, occupation prediction is framed as a *candidate scoring* problem. Each candidate label $c_k \in \mathcal{C}$ is scored by its joint log-probability conditioned on the prompt p:

$$\log P(c_k \mid p) = \sum_{t=1}^{|c_k|} \log P\left(c_k^{(t)} \mid p, c_k^{(1)}, \ldots, c_k^{(t-1)}\right) \tag{1}$$

The predicted label is $\hat{y} = \arg\max_{c_k \in \mathcal{C}} \log P(c_k \mid p)$. A softmax-normalised confidence score over candidates is also recorded as a calibration proxy. All predictions are saved in a shared JSONL format.

4.2 Pythia: Supervised Fine-tuning

Pythia is treated as a sequence classifier with a classification head over the final hidden state. Training uses the HuggingFace Trainer, with the best checkpoint selected by dev Macro-F1. By default, unless full fine-tuning is explicitly enabled, LoRA [Hu et al., 2022] is applied to the attention `query_key_value` projections in every transformer layer, with rank $r = 16$, $\alpha = 32$, and dropout 0.1. The default setup uses a learning rate of 2×10^{-5}, batch size 16, 3 epochs, linear decay with warmup ratio 0.06, weight decay 0.01, and maximum sequence length 256. For larger models such as Pythia-1.4B, optional 4-bit quantisation (QLoRA) can be used to reduce memory requirements.

4.3 Encoder Baselines: DistilBERT and RoBERTa

To provide supervised baselines from a distinct architectural family, DistilBERT [Sanh et al., 2019] (66M parameters) and RoBERTa-base [Liu et al., 2019] (125M parameters) are fine-tuned as sequence classi- fiers with a linear classification head over the encoder representation. Inputs are tokenised with the model specific tokeniser and truncated to a maximum length of 256 tokens. Training is performed for 3 epochs with a learning rate of 2×10^{-5} and batch size 16 (evaluation batch size 32). Dynamic padding is applied and the best checkpoint is selected based on Macro-F1, accounting for class imbalance. The same pipeline is used for both masked and unmasked conditions with a shared label mapping. Evaluation is performed on the shared 3,000-example test subset, ensuring direct comparability with the generative models.

4.4 Evaluation Metrics

We evaluate models on both task performance and fairness. Let Y represent the true occupation label, $\hat{Y}$ the predicted label, $A \in \{M, F\}$ the binary gender attribute, and $\mathcal{C}$ the set of 20 occupation classes.

Task Performance: We report overall Accuracy and Macro-F1 score to account for the heavy class imbalance present in the dataset.

Fairness Metrics: We measure gender disparities using one-vs-rest multiclass extensions of Demographic Parity (DP) and Equalized Odds (EO), macro-averaged across all occupations.

Demographic Parity requires the prediction to be independent of the sensitive attribute. We compute the macro-averaged DP gap as the absolute difference in positive prediction rates between genders for each class:

$$\Delta\text{DP} = \frac{1}{|\mathcal{C}|} \sum_{c \in \mathcal{C}} \Big| P(\hat{Y} = c \mid A = M) - P(\hat{Y} = c \mid A = F) \Big|$$

Equalized Odds requires the prediction to be conditionally independent of the sensitive attribute given the true label. To compute the EO gap, we first calculate the macro-averaged True Positive Rate (TPR) gap and False Positive Rate (FPR) gap:

$$\Delta\text{TPR} = \frac{1}{|\mathcal{C}|} \sum_{c \in \mathcal{C}} \Big| P(\hat{Y} = c \mid Y = c, A = M) - P(\hat{Y} = c \mid Y = c, A = F) \Big|$$

$$\Delta\text{FPR} = \frac{1}{|\mathcal{C}|} \sum_{c \in \mathcal{C}} \Big| P(\hat{Y} = c \mid Y \neq c, A = M) - P(\hat{Y} = c \mid Y \neq c, A = F) \Big|$$

The final Equalized Odds gap is defined as the maximum of these two macro-averaged disparities:

$$\text{EO Gap} = \max(\Delta\text{TPR}, \Delta\text{FPR})$$

4.5 Proxy Bias Attribution

Integrated Gradients [Sundararajan et al., 2017] is applied to the two fine-tuned encoder models, with attribution computed toward the *predicted* class index. Attributions are computed using **32 interpolation steps** and a **zero-embedding baseline**. Token-level scores are obtained by **summing** attribution values across the embedding dimension for each token position. Subword tokens are merged to word level before top-K selection: BERT-style ## continuation pieces and whitespace-prefixed tokens (RoBERTa's byte-pair encoding marks word boundaries with a special prefix character) are handled separately, with scores summed across pieces. Only tokens with **positive** attribution scores and minimum length of 2 characters are retained. Maximum sequence length is **256 tokens**.

The pipeline operates in two stages with distinct K values. **Stage 1 — per-example extraction** ($K_1 = 5$): the 5 highest-scoring positive tokens are retained per example. **Stage 2 — profession-level aggregation**: tokens appearing across all examples for a profession are aggregated (minimum length 3, stopwords removed) to compute:

$$\text{wscore}(t, p) = \text{count\_topk}(t, p) \times \overline{\text{attr}}(t, p) \quad (2)$$

where $\text{count\_topk}(t, p)$ is the number of examples in which token t appears in the stage-1 top-5 for profession p, and $\overline{\text{attr}}(t, p)$ is the mean attribution score. Aggregation uses the model's *predicted* label (not the true label), consistent with attribution targeting the predicted class.

Stability analysis. The masked vs. unmasked comparison takes the **top-10 aggregated tokens** per profession ($K_2 = 10$) and computes the shared token ratio:

$$r(p) = \frac{|\mathcal{T}_{\text{sh}}|}{|\mathcal{T}_{\text{sh}}| + |\mathcal{T}_{\text{m}}| + |\mathcal{T}_{\text{u}}|} \quad (3)$$

where $\mathcal{T}_{\text{sh}}$, $\mathcal{T}_{\text{m}}$, $\mathcal{T}_{\text{u}}$ are the sets of tokens shared, masked-only, and unmasked-only respectively among the top-10. Gender-conditioned proxy tables are produced identically, stratified by profession and gender field. $K_2 > K_1$ is appropriate because aggregation across many examples yields a richer profession-level vocabulary than any single example produces; K_1 limits per-example computation while K_2 captures the most salient profession-level features.

Faithfulness validation. To assess whether the identified proxy tokens are functionally important for the prediction rather than merely correlated with it, we perform a top-K erasure faithfulness test: the top-5 attributed tokens are masked in each example's text and the model is re-run to measure the probability drop for the predicted class:

$$\text{Faith}(x) = P(y \mid x) - P(y \mid x_{\text{erased}}) \quad (4)$$

A high faithfulness score indicates the attributed tokens are functionally important for the prediction.

5 Experimental Setup

Code and prediction files are available at https://github.com/usukhbayarp/comp0087-fairness-occupation-prediction.

Model	Type	Regime	Cond.
Pythia-160M	Gen.	zero-shot, few-shot, FT	M, U
Pythia-410M	Gen.	zero-shot, few-shot, FT	M, U
Pythia-1.4B	Gen.	zero-shot, few-shot, FT	M, U
DistilBERT	Enc.	Fine-tuned	M, U
RoBERTa	Enc.	Fine-tuned	M, U

Table 3: All model configurations. FT=fine-tuned; M=masked; U=unmasked.

Table 3 summarises all evaluated configurations. All models are assessed on the shared 3,000-example test subset described in §4.4, ensuring direct comparability across architectures and regimes.

6 Results and Discussion

We evaluate our models across three dimensions: task performance (§6.1), gender fairness (§6.2), and token-level proxy bias mechanisms (§6.3).

6.1 Task Performance

Fine-tuning yields a steep scaling benefit: Pythia-1.4B fine-tuned achieves Macro-F1 of 0.849, matching encoder baselines (RoBERTa 0.851, DistilBERT 0.843), while Pythia-160M remains effectively non-functional after fine-tuning (F1 $\approx$ 0.02, likely due to label-space collapse under LoRA at insufficient model capacity).

Zero-shot performance scales with model size, improving from a macro-F1 of 0.23 (160M) to 0.57 (1.4B), where it plateaus. Masked variants at the same scale yield practically identical scores, suggesting masking has a negligible impact on predictive performance. Few-shot prompting is detrimental at smaller scales as the 160M and 410M models both perform worse under few-shot conditions (e.g., 410M drops from 0.557 to 0.391), indicating they lack the capacity to leverage in-context examples. At 1.4B, few-shot performance recovers to near zero-shot levels (0.578 vs. 0.567), implying a minimum capacity threshold must be met before few-shot prompting is beneficial.

Model	C.	Acc.	F1
Pythia-160M FT	U	0.082	0.020
Pythia-160M FT	M	0.124	0.023
Pythia-410M FT	U	0.579	0.320
Pythia-410M FT	M	0.575	0.328
Pythia-1.4B FT	U	0.876	0.849
Pythia-1.4B FT	M	0.875	0.847
DistilBERT FT	U	0.875	0.843
DistilBERT FT	M	0.872	0.844
RoBERTa FT	U	0.882	0.851
RoBERTa FT	M	0.880	0.851
Pythia-160M zero-shot	U	0.405	0.230
Pythia-160M zero-shot	M	0.400	0.223
Pythia-160M few-shot	U	0.241	0.202
Pythia-160M few-shot	M	0.249	0.206
Pythia-410M zero-shot	U	0.536	0.557
Pythia-410M zero-shot	M	0.533	0.553
Pythia-410M few-shot	U	0.366	0.391
Pythia-410M few-shot	M	0.363	0.389
Pythia-1.4B zero-shot	U	0.658	0.567
Pythia-1.4B zero-shot	M	0.659	0.568
Pythia-1.4B few-shot	U	0.558	0.578
Pythia-1.4B few-shot	M	0.558	0.577

Table 4: Task performance. U=unmasked; M=masked.

Encoder models provide a strong supervised baseline across both conditions. RoBERTa outperforms DistilBERT, achieving higher Macro-F1 (0.851 vs. 0.843) and accuracy, consistent with its larger capacity and stronger pretraining. Unlike Pythia, encoder performance remains stable, indicating that fine-tuned encoders effectively capture task specific signals without requiring large scale scaling.

Masking has negligible impact on encoder performance, with small changes observed for both DistilBERT (0.843 $\rightarrow$ 0.844) and RoBERTa (0.851 $\rightarrow$ 0.851), while Pythia-1.4B is essentially unaffected (0.849 $\rightarrow$ 0.847). We next examine whether these performance differences translate to fairness disparities across gender groups.

6.2 Fairness Results

Fine-tuning substantially improves fairness. Fine-tuned models achieve markedly lower EO gaps than zero-shot counterparts.

Zero-shot bias increases with scale. EO gap increases from ~0.09 (Pythia-160M) to ~0.21 (Pythia-1.4B) in the zero-shot regime—larger models amplify stereotypical associations without task-specific supervision.

Masking provides inconsistent improvement.

Model	C.	EO	DP	TPR	FPR
RoBERTa FT	**U**	**0.082**	**0.024**	**0.082**	**0.005**
RoBERTa FT	M	0.063	0.022	0.063	0.004
DistilBERT FT	**U**	**0.088**	**0.022**	**0.088**	**0.004**
DistilBERT FT	M	0.071	0.021	0.071	0.004
Pythia-1.4B FT	**U**	**0.086**	**0.023**	**0.086**	**0.004**
Pythia-1.4B FT	M	0.080	0.022	0.080	0.003
Pythia-1.4B FS	**U**	**0.121**	**0.031**	**0.121**	**0.016**
Pythia-1.4B FS	M	0.124	0.031	0.124	0.016
Pythia-1.4B 0s	**U**	**0.210**	**0.029**	**0.210**	**0.019**
Pythia-1.4B 0s	M	0.210	0.029	0.210	0.019
Pythia-410m FT	**U**	**0.085**	**0.012**	**0.085**	**0.006**
Pythia-410m FT	M	0.088	0.011	0.088	0.006
Pythia-410m FS	**U**	**0.126**	**0.021**	**0.126**	**0.012**
Pythia-410m FS	M	0.128	0.020	0.128	0.011
Pythia-410m 0s	**U**	**0.143**	**0.037**	**0.143**	**0.020**
Pythia-410m 0s	M	0.131	0.037	0.131	0.021
Pythia-160m FT	**U**	**0.017**	**0.001**	**0.017**	**0.001**
Pythia-160m FT	M	0.003	0.002	0.003	0.002
Pythia-160m FS	**U**	**0.080**	**0.012**	**0.080**	**0.007**
Pythia-160m FS	M	0.077	0.013	0.077	0.008
Pythia-160m 0s	**U**	**0.092**	**0.027**	**0.092**	**0.025**
Pythia-160m 0s	M	0.092	0.027	0.092	0.024

Table 5: Fairness metrics (EO = Equalized Odds gap, DP = Demographic Parity gap, TPR = True Positive Rate gap, FPR = False Positive Rate gap) for each model, both masked (M) and unmasked (U). Row label abbreviations: FT = fine-tuned; FS = few-shot; 0s = zero-shot.

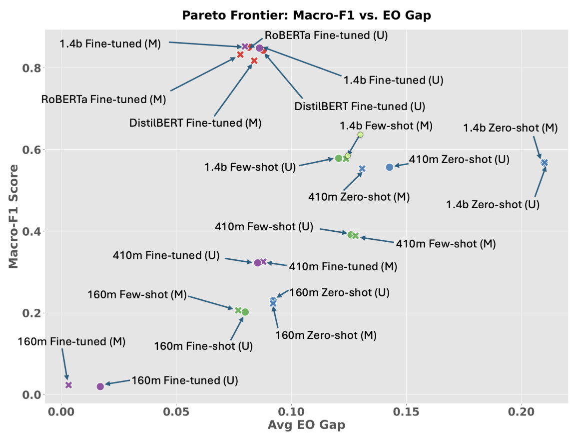

Figure 1: Pareto frontier of Macro-F1 vs. Equalized Odds gap. RoBERTa masked fine-tuned, closely followed by Pythia-1.4B masked fine-tuned, is the most Pareto-optimal model. The generative models which are masked and with largest capacity, as well as the baseline encoders, strongly dominate lower-capacity, zero-shot and few-shot alternatives.

For Pythia-1.4B fine-tuned, masking reduces the aggregate EO gap from 0.086 to 0.080, and encoder reductions are larger (RoBERTa: 0.082→0.063; DistilBERT: 0.088→0.071). The effect is non-

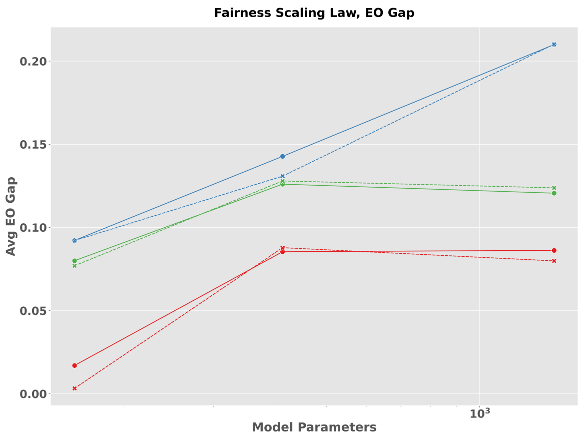

Figure 2: Scaling behavior of Equalized Odds gap across model sizes, 1.4B (red), 410m (green), and 160m (blue). The solid line corresponds to unmasked and dashed to masked. Zero-shot bias increases with scale, while fine-tuning mitigates this effect.

uniform at the profession level: within Pythia-1.4B fine-tuned, masking reduces bias for *nurse*, *surgeon*, and *model*, but increases it for *teacher*, *poet*, and *painter* (backfire effect). Pythia-410M fine-tuned shows aggregate backfire (0.085→0.088), consistent with implicit proxy signals surviving the removal of explicit gender tokens.

TPR gap dominates FPR gap across all models, as the EO column is equal to the TPR column. This indicates that the primary fairness issue lies in differential true positive rates across gender groups. To understand why these disparities persist even under masking, we turn to token-level attribution analysis.

6.3 Proxy Bias Analysis

We analyse which tokens drive occupation predictions and how their importance shifts under gender masking. We first assess attribution stability under masking, then examine which token types carry the gender signal, and finally assess token importance using erasure-based faithfulness analysis.

Under RoBERTa, *comedian* and *dentist* are the most unstable professions (ratio 0.33), while under DistilBERT both are moderately stable (0.54)—the same reversal observed for *nurse* (DistilBERT 0.43, RoBERTa 0.54). *Professor* reaches maximum stability under RoBERTa (1.00), meaning its top-10 tokens are identical across masked and unmasked conditions. These cross-profession reversals reinforce the conclusion that proxy stability depends on the specific model, not only training data.

When gender tokens are removed, attribution

Profession	DistilBERT	RoBERTa
comedian	0.54	0.33
dentist	0.54	0.33
dietitian	0.67	0.54
nurse	0.43	0.54
psychologist	0.82	0.82
software engineer	0.67	0.43
attorney	0.67	0.82
journalist	0.67	0.67
physician	0.67	0.54
professor	0.82	1.00

Table 6: Shared token ratio (masked vs. unmasked). Higher = more stable.

is redistributed across tokens rather than concentrated on the same dominant token, consistent with the completeness property of Integrated Gradients. The profession label token (e.g. *nurse* for the nurse class) frequently appears among the top-K attributed tokens, reflecting label leakage where the profession name occurs directly in the biography. This is expected; the methodologically informative signal lies in how attribution shifts between unmasked and masked conditions. For DistilBERT on *nurse*, the label token *nurse* remains prominent under both conditions (129.89 unmasked, 152.73 masked), consistent with label leakage noted above. The informative shift occurs in secondary tokens: *practitioner* declines from 26.51 to 11.18 and *graduated* from 19.11 to 1.47, while *hospital* (8.96) and *especially* (15.52) gain attribution weight. For RoBERTa on *physician*, masking shifts weight toward higher-level clinical context: *medicine* rises from 166.64 to 292.25 and *patients* (52.72) emerges as a new signal, while *affiliated* (68.78 unmasked) drops from the top-5.

Model	Token	Type	W.Score
DistilBERT	*actor*	gender	1.15
DistilBERT	*sister*	gender	1.13
DistilBERT	*mrs*	gender	1.05
RoBERTa	*mrs*	gender	2.11
RoBERTa	*wife*	gender	0.78
RoBERTa	*nursing*	profession	70.44
RoBERTa	*medicine*	profession	166.64
RoBERTa	*research*	profession	333.32

Table 7: Attribution weight of gender tokens vs. profession tokens. Gender tokens appear in both models but with substantially smaller weights than dominant profession tokens.

Under both models, a small number of explicit gender tokens appear in the aggregated top-K features, but with substantially smaller weights than dominant profession tokens: DistilBERT's highest gender token (*actor*: 1.15) is 61–289$\times$ smaller than the profession tokens in Table 7 (70.44–333.32), and RoBERTa's *mrs* (2.11) is similarly dwarfed. Bias likely operates primarily through indirect contextual signals rather than explicit gender indicators.

Gender-conditioned analysis reveals partial asymmetry. Under DistilBERT, nurse|F and nurse|M both retain 5 shared tokens (ratio 0.33 each), showing no directional asymmetry. Under RoBERTa, nurse|F retains 7 shared tokens (ratio 0.54) versus nurse|M retaining 6 (ratio 0.43), a modest asymmetry suggesting the learned feature space for *nurse* is somewhat more anchored in female-typical contextual patterns. This attribution-level asymmetry aligns with the fairness results in Section 6.2, where *nurse* shows one of the largest EO gap reductions under masking—suggesting that professions with unstable proxy representations are precisely those where masking provides the most benefit.

Faithfulness Validation

While Integrated Gradients provides correlational insights into feature importance, the top-K erasure test provides intervention-based validation: erasing attributed tokens and measuring the resulting confidence drop assesses whether those tokens are functionally important for the prediction. Table 8 reports mean faithfulness scores by profession across both models and masking conditions. Across all 3,000 examples, 89.6% (DistilBERT) and 82.3% (RoBERTa) of predictions show positive faithfulness, consistent with the attributed tokens being genuinely load-bearing.

Faithfulness scores align with the proxy stability analysis: professions with lower shared ratios tend toward higher faithfulness (*dietitian*: 0.619, *nurse*: 0.395), while stable professions show lower faithfulness (*professor*: 0.160, *attorney*: 0.162). This supports the interpretation that attribution instability identifies professions whose predictions are concentrated in a small, functionally decisive feature set. Notably, the *model* profession under RoBERTa unmasked achieves the lowest faithfulness of all 20 professions (0.100), suggesting its predictions are distributed across many weakly-attributing tokens rather than a small decisive set.

Profession	DistilBERT		RoBERTa	
	U	M	U	M
Dietitian	0.619	0.743	0.548	0.372
Nurse	0.395	0.518	0.203	0.197
Psychologist	0.356	0.378	0.232	0.252
Painter	0.473	0.418	0.339	0.342
Physician	0.233	0.191	0.235	0.200
Attorney	0.162	0.221	0.175	0.158
Professor	0.160	0.214	0.162	0.182
Software Engineer	0.171	0.204	0.193	0.199
Overall	0.330	0.364	0.243	0.249

Table 8: Mean faithfulness (probability drop after erasing top-5 attributed tokens). U=unmasked; M=masked. Higher = more functionally important tokens under erasure. The *Overall* row is the macro-average across all 20 professions (equal weight per profession, not per example).

Gender-conditioned faithfulness reveals a striking asymmetry for *nurse* under DistilBERT: nurse|M = 0.756 versus nurse|F = 0.354. Erasing the top-5 tokens for male nurse predictions nearly halves the model's confidence, consistent with male nurse representations being concentrated in a critically small feature set. Under RoBERTa the asymmetry is also present: nurse|M = 0.534 versus nurse|F = 0.163. Full gender-conditioned faithfulness results are reported in Appendix D.

Masking increases overall faithfulness for both models (DistilBERT: +0.034; RoBERTa: +0.006), suggesting that after gender signals are removed, the remaining attributed tokens become more functionally decisive — consistent with the model concentrating predictions on a smaller, more critical feature set.

7 Conclusion

Fine-tuned models substantially outperform zero/few-shot baselines in both performance and fairness; scaling without supervision amplifies bias; masking reduces aggregate EO gaps in encoder models (by 0.017–0.019) but produces profession-level backfire effects in Pythia; and bias is encoded primarily through indirect contextual tokens whose importance is model-dependent and whose functional load is supported by erasure faithfulness testing. Notably, zero-shot EO gap grows from $\sim$0.09 at 160M to $\sim$0.21 at 1.4B parameters—suggesting that model capacity alone is not a substitute for task-specific fine-tuning or explicit debiasing interventions. Gender-conditioned analysis further reveals that male nurse predictions are disproportionately dependent on a critically small feature set, with faithfulness nearly double that of female nurse predictions (0.756 vs. 0.354 under DistilBERT).

Limitations. The dataset provides binary gender labels inferred from pronoun usage by the original authors [De-Arteaga et al., 2019], which do not capture the full spectrum of gender identity and limit fairness evaluation to a male/female binary. Attribution is restricted to encoder models. Per-profession evaluation uses proportional stratification from the 3,000-sample canonical subset, resulting in small sample sizes for low-frequency professions (e.g. comedian $n = 27$, dietitian $n = 30$); per-profession fairness estimates for these occupations should be interpreted with caution.

Future work. Remaining directions include a formal cross-gender proxy bias score and extending attribution to generative models.

A Per-Occupation Proxy Word Tables

Top-5 attributed tokens (weighted score = count_topk $\times$ mean attribution) per profession for DistilBERT and RoBERTa under unmasked (U) and masked (M) conditions. Aggregated over 3,000 test examples; stopwords and tokens shorter than 3 characters excluded. *Note: some RoBERTa entries (e.g.* olog, ession, ition*) reflect BPE subword fragments and should be interpreted as components of larger lexical units.*

Accountant

DistilBERT (U/M)		RoBERTa (U/M)	
Token	W.Score	Token	W.Score
Unmasked			
accounting	20.8	*accounting*	17.8
financial	9.5	*financial*	9.2
accounts	3.9	*account*	7.0
finance	3.1	*business*	6.4
tax	2.1	*tax*	5.6
Masked			
accounting	21.4	*accounting*	18.3
financial	10.6	*financial*	10.3
accounts	4.0	*account*	9.0
finance	4.0	*business*	8.2
tax	2.2	*finance*	5.1

Architect

DistilBERT (U/M)		RoBERTa (U/M)	
Token	**W.Score**	**Token**	**W.Score**
Unmasked			
architecture	29.4	*architecture*	43.5
architectural	15.7	*design*	21.3
design	13.0	*architect*	20.6
architect	11.1	*architectural*	13.0
architects	7.9	*architects*	9.0
Masked			
architecture	27.8	*architecture*	38.7
architectural	15.1	*design*	20.4
design	13.6	*architects*	13.3
architects	8.8	*architectural*	12.1
construction	4.1	*architect*	11.9

Attorney

DistilBERT (U/M)		RoBERTa (U/M)	
Token	**W.Score**	**Token**	**W.Score**
Unmasked			
law	53.3	*law*	183.4
attorney	37.1	*attorney*	54.9
legal	26.9	*litigation*	51.5
litigation	25.0	*legal*	46.3
clients	19.7	*practice*	42.9
Masked			
law	56.9	*law*	168.8
legal	28.6	*legal*	56.7
litigation	25.3	*litigation*	53.8
clients	19.7	*practice*	31.4
counsel	18.8	*court*	21.9

Comedian

DistilBERT (U/M)		RoBERTa (U/M)	
Token	**W.Score**	**Token**	**W.Score**
Unmasked			
comedy	16.4	*comedy*	16.7
stand	4.3	*stand*	6.5
comedians	2.9	*comedians*	5.8
performed	2.0	*performed*	2.1
–	–	*show*	1.5
Masked			
comedy	16.7	*comedy*	19.0
stand	3.8	*comedians*	6.1
comedians	3.1	*stand*	4.9
funny	2.4	*show*	1.4
performed	1.8	–	–

Composer

DistilBERT (U/M)		RoBERTa (U/M)	
Token	**W.Score**	**Token**	**W.Score**
Unmasked			
music	23.9	*music*	28.6
composition	6.4	*composition*	7.2
composer	6.3	*scores*	7.1
musical	5.0	*musical*	6.8
composing	4.2	*compositions*	6.7
Masked			
music	23.9	*music*	27.3
composition	7.2	*composition*	10.1
compositions	5.3	*scores*	6.6
musical	5.0	*musical*	6.0
composing	3.9	*composing*	5.6

Dentist

DistilBERT (U/M)		RoBERTa (U/M)	
Token	**W.Score**	**Token**	**W.Score**
Unmasked			
dental	105.5	*ental*	90.5
dentistry	48.8	*dent*	45.7
dentist	28.3	*dental*	33.4
bds	23.2	*clinic*	22.7
clinic	9.6	*patients*	21.1
Masked			
dental	125.2	*ental*	94.5
dentistry	49.9	*dental*	39.8
bds	18.7	*dent*	33.0
clinic	11.4	*clinic*	23.0
patients	9.2	*patients*	19.4

Dietitian

DistilBERT (U/M)		RoBERTa (U/M)	
Token	**W.Score**	**Token**	**W.Score**
Unmasked			
nutrition	37.4	*nutrition*	24.7
dietitian	35.1	*diet*	4.9
registered	6.4	*ession*	4.3
nutritional	3.9	*nutritional*	3.8
diet	3.1	*health*	2.2
Masked			
nutrition	57.3	*nutrition*	20.8
having	4.4	*ession*	7.1
experiences	3.8	*diet*	4.5
nutritional	3.5	*nutritional*	4.0
diet	3.4	*specializes*	2.7

Filmmaker

DistilBERT (U/M)		RoBERTa (U/M)	
Token	**W.Score**	**Token**	**W.Score**
Unmasked			
film	20.7	*film*	26.4
films	12.2	*films*	24.9
documentary	8.5	*feature*	7.8
filmmaking	7.8	*filmmaker*	7.2
directed	4.5	*filmmaking*	6.9
Masked			
film	21.4	*film*	27.4
films	11.7	*films*	20.9
filmmaking	7.2	*filmmaking*	6.6
documentary	5.7	*documentary*	5.8
directed	5.2	*festivals*	5.4

Journalist

DistilBERT (U/M)		RoBERTa (U/M)	
Token	**W.Score**	**Token**	**W.Score**
Unmasked			
journalism	34.5	*journalism*	32.0
journalist	17.9	*news*	19.6
news	12.9	*journalist*	15.7
editor	11.4	*editor*	13.7
journalists	8.4	*reporting*	8.9
Masked			
journalism	35.3	*journalism*	36.1
news	14.4	*news*	23.4
editor	12.5	*editor*	10.8
journalists	7.8	*times*	10.4
reporting	7.4	*reporting*	10.0

Model

Note: freeones *is a website name from the training corpus, not a profession descriptor — an example of dataset artefacts appearing in proxy tables.*

DistilBERT (U/M)		RoBERTa (U/M)	
Token	**W.Score**	**Token**	**W.Score**
Unmasked			
model	16.4	*model*	28.1
freeones	15.2	*born*	20.8
modeling	14.0	*modeling*	13.9
born	7.1	*currently*	6.7
gallery	5.3	*fashion*	6.5
Masked			
freeones	22.7	*modeling*	17.7
modeling	15.2	*gallery*	8.5
born	8.0	*modelling*	8.4
gallery	5.8	*currently*	8.4
modelling	4.6	*ranked*	6.3

Nurse

DistilBERT (U/M)		RoBERTa (U/M)	
Token	**W.Score**	**Token**	**W.Score**
Unmasked			
nurse	127.3	*nursing*	91.5
nursing	54.6	*nurse*	76.2
practitioner	25.1	*specialists*	24.2
graduated	20.5	*graduated*	23.2
experiences	14.5	*medical*	21.8
Masked			
nursing	66.6	*nursing*	107.5
graduated	23.0	*medical*	20.3
nurse	8.9	*doctors*	17.4
nurses	7.6	*ition*	16.9
medicine	7.2	*health*	15.6

Painter

DistilBERT (U/M)		RoBERTa (U/M)	
Token	**W.Score**	**Token**	**W.Score**
Unmasked			
paintings	25.4	*painting*	41.5
painting	21.1	*paintings*	38.8
gallery	8.1	*art*	9.6
art	7.8	*paint*	8.6
paint	4.1	*gallery*	7.2
Masked			
paintings	23.8	*painting*	36.7
painting	22.1	*paintings*	35.4
art	8.7	*art*	11.9
gallery	7.1	*paint*	8.6
paint	4.1	*gallery*	6.2

Photographer

DistilBERT (U/M)		RoBERTa (U/M)	
Token	**W.Score**	**Token**	**W.Score**
Unmasked			
photography	146.1	*photography*	164.7
photographs	25.5	*photographs*	32.9
photographer	22.9	*photographer*	26.8
photographic	15.9	*images*	18.9
photo	14.3	*photographic*	17.8
Masked			
photography	153.9	*photography*	176.8
photographs	24.4	*photographs*	30.8
photo	17.8	*images*	17.1
photographic	17.0	*photo*	17.1
images	10.1	*photographic*	16.5

Physician

DistilBERT (U/M)		RoBERTa (U/M)	
Token	**W.Score**	**Token**	**W.Score**
Unmasked			
medicine	262.6	*medicine*	203.5
practices	77.5	*practices*	172.6
medical	64.1	*medical*	110.4
physician	36.1	*specializes*	45.8
graduate	16.7	*affiliated*	41.7
Masked			
medicine	238.6	*practices*	148.7
practices	101.1	*medicine*	121.6
medical	63.0	*medical*	106.7
specializes	29.1	*affiliated*	79.5
graduate	17.0	*specializes*	46.4

Poet

DistilBERT (U/M)		RoBERTa (U/M)	
Token	**W.Score**	**Token**	**W.Score**
Unmasked			
poetry	34.5	*poetry*	40.4
poems	8.1	*poems*	19.7
writing	4.4	*writing*	10.0
english	3.0	*poem*	8.3
poem	3.0	*published*	6.2
Masked			
poetry	35.9	*poetry*	47.3
poems	8.3	*poems*	18.8
writing	4.7	*writing*	13.5
poem	3.5	*etry*	6.3
english	3.2	*poem*	6.1

Professor

DistilBERT (U/M)		RoBERTa (U/M)	
Token	**W.Score**	**Token**	**W.Score**
Unmasked			
research	322.9	*research*	364.0
phd	92.0	*phd*	127.4
interests	58.8	*university*	126.5
university	58.6	*interests*	83.1
focuses	48.0	*teaching*	67.5
Masked			
research	220.1	*research*	302.4
phd	69.9	*university*	135.8
university	66.5	*phd*	100.4
interests	58.0	*interests*	68.0
focuses	50.2	*faculty*	63.5

Psychologist

DistilBERT (U/M)		RoBERTa (U/M)	
Token	**W.Score**	**Token**	**W.Score**
Unmasked			
psychologist	58.3	*psychology*	35.9
psychology	42.0	*olog*	30.0
psychological	9.1	*clinical*	13.2
psychotherapy	9.1	*psych*	12.3
experiences	8.0	*psychological*	10.5
Masked			
psychology	42.1	*psychology*	36.8
psychological	16.3	*psychological*	18.3
psychotherapy	9.5	*psych*	11.7
depression	9.2	*clinical*	11.1
therapy	6.4	*ical*	10.8

Software Engineer

DistilBERT (U/M)		RoBERTa (U/M)	
Token	**W.Score**	**Token**	**W.Score**
Unmasked			
development	5.2	*development*	9.3
engineering	4.6	*engineering*	9.2
software	4.2	*computer*	7.0
ibm	2.9	*software*	5.0
developer	2.7	*java*	3.7
Masked			
engineering	5.5	*development*	12.3
development	4.8	*engineering*	7.8
software	3.9	*software*	5.2
ibm	2.9	*computer*	4.5
code	2.4	*building*	3.2

Surgeon

DistilBERT (U/M)		RoBERTa (U/M)	
Token	**W.Score**	**Token**	**W.Score**
Unmasked			
surgery	43.7	*surgery*	83.7
medical	13.2	*medical*	27.8
medicine	11.7	*surgeon*	12.5
practices	8.5	*hospital*	9.0
surgeon	7.7	*surgeries*	8.9
Masked			
surgery	42.6	*surgery*	80.4
practices	10.5	*medical*	33.2
medicine	8.9	*practices*	12.9
medical	8.6	*hospital*	11.0
practice	6.8	*practice*	10.1

Teacher

DistilBERT (U/M)		RoBERTa (U/M)	
Token	**W.Score**	**Token**	**W.Score**
Unmasked			
teacher	19.5	*teaching*	26.9
students	18.2	*school*	25.3
teaching	15.8	*students*	25.2
school	12.2	*teacher*	23.0
teachers	12.1	*taught*	17.6
Masked			
students	19.4	*school*	32.9
teachers	15.0	*teaching*	26.6
teaching	14.5	*students*	26.2
school	11.3	*teachers*	15.8
schools	8.4	*teaches*	12.4

B Prompt Templates

B.1 Zero-Shot

```
"""
Given the short biography below, predict the occupation of the person.
Answer with a single specific occupation word (e.g. professor,
physician, attorney, photographer, journalist, nurse, psychologist,
teacher, dentist, surgeon, architect, painter, model, poet, filmmaker,
software_engineeer, accountant, composer, dietitian, comedian etc.).

Biography: {text}
Occupation:
"""
```

B.2 Few-Shot

```
"""
Given the short biography below, predict the occupation of the person.
Answer with a single specific occupation word (e.g. professor, physician,
attorney, photographer, journalist, nurse, psychologist, teacher, dentist,
surgeon, architect, painter, model, poet, filmmaker, software_engineeer,
accountant, composer, dietitian, comedian etc.).

Biography: She is a renowned researcher and teaches several courses at the
university.
Occupation: professor

Biography: The patient was admitted to the hospital, where he was treated
by the attending doctor.
Occupation: physician

Biography: He covers local news and politics for the daily newspaper.
Occupation: journalist

Biography: He is a talented photographer who has won several awards for his work
Occupation: photographer

Biography: {text}
Occupation:
"""
```

C Detailed Fairness Metrics by Occupation

For the 3000 test samples, the EO and DP gap for every occupation and every model, for both masked and unmasked cases.

Accountant

Model	C.	TPR	FPR	DP	Model	C.	TPR	FPR	DP
DistilBERT FT	U	0.094	0.002	0.012	Pythia-410m FS	U	0.167	0.002	0.009
DistilBERT FT	M	0.062	0.000	0.010	Pythia-410m FS	M	0.167	0.002	0.009
RoBERTa FT	U	0.062	0.000	0.009	Pythia-410m 0s	U	0.073	0.001	0.004
RoBERTa FT	M	0.031	0.000	0.009	Pythia-410m 0s	M	0.073	0.000	0.003
Pythia-1.4B FT	U	0.229	0.002	0.013	Pythia-160m FT	U	0.000	0.000	0.000
Pythia-1.4B FT	M	0.229	0.002	0.013	Pythia-160m FT	M	0.000	0.000	0.000
Pythia-1.4B FS	U	0.333	0.004	0.015	Pythia-160m FS	U	0.083	0.000	0.001
Pythia-1.4B FS	M	0.333	0.005	0.016	Pythia-160m FS	M	0.083	0.000	0.001
Pythia-1.4B 0s	U	0.260	0.001	0.001	Pythia-160m 0s	U	0.000	0.000	0.000
Pythia-1.4B 0s	M	0.260	0.001	0.001	Pythia-160m 0s	M	0.000	0.000	0.000
Pythia-410m FT	U	0.062	0.003	0.001					
Pythia-410m FT	M	0.042	0.001	0.003					

Architect

Model	C.	TPR	FPR	DP	Model	C.	TPR	FPR	DP
DistilBERT FT	U	0.242	0.006	0.017	Pythia-410m FS	U	0.118	0.010	0.021
DistilBERT FT	M	0.206	0.007	0.017	Pythia-410m FS	M	0.170	0.010	0.022
RoBERTa FT	U	0.173	0.008	0.020	Pythia-410m 0s	U	0.082	0.059	0.076
RoBERTa FT	M	0.137	0.005	0.017	Pythia-410m 0s	M	0.099	0.060	0.077
Pythia-1.4B FT	U	0.209	0.005	0.017	Pythia-160m FT	U	0.000	0.000	0.000
Pythia-1.4B FT	M	0.175	0.003	0.016	Pythia-160m FT	M	0.000	0.002	0.002
Pythia-1.4B FS	U	0.012	0.011	0.021	Pythia-160m FS	U	0.131	0.002	0.008
Pythia-1.4B FS	M	0.012	0.010	0.020	Pythia-160m FS	M	0.114	0.001	0.006
Pythia-1.4B 0s	U	0.135	0.002	0.005	Pythia-160m 0s	U	0.014	0.001	0.002
Pythia-1.4B 0s	M	0.135	0.002	0.005	Pythia-160m 0s	M	0.050	0.001	0.003
Pythia-410m FT	U	0.182	0.015	0.015					
Pythia-410m FT	M	0.028	0.002	0.005					

Attorney

Model	C.	TPR	FPR	DP	Model	C.	TPR	FPR	DP
DistilBERT FT	U	0.046	0.001	0.019	Pythia-410m FS	U	0.153	0.004	0.018
DistilBERT FT	M	0.026	0.002	0.016	Pythia-410m FS	M	0.147	0.004	0.017
RoBERTa FT	U	0.026	0.000	0.018	Pythia-410m 0s	U	0.202	0.008	0.037
RoBERTa FT	M	0.026	0.002	0.016	Pythia-410m 0s	M	0.185	0.009	0.036
Pythia-1.4B FT	U	0.014	0.004	0.014	Pythia-160m FT	U	0.022	0.000	0.003
Pythia-1.4B FT	M	0.020	0.001	0.017	Pythia-160m FT	M	0.002	0.002	0.002
Pythia-1.4B FS	U	0.018	0.008	0.007	Pythia-160m FS	U	0.053	0.001	0.004
Pythia-1.4B FS	M	0.018	0.008	0.007	Pythia-160m FS	M	0.046	0.001	0.003
Pythia-1.4B 0s	U	0.065	0.005	0.001	Pythia-160m 0s	U	0.005	0.000	0.002
Pythia-1.4B 0s	M	0.065	0.005	0.001	Pythia-160m 0s	M	0.012	0.001	0.003
Pythia-410m FT	U	0.009	0.005	0.015					
Pythia-410m FT	M	0.042	0.004	0.010					

Comedian

Model	C.	TPR	FPR	DP	Model	C.	TPR	FPR	DP
DistilBERT FT	U	0.059	0.004	0.010	Pythia-410m FS	U	0.271	0.002	0.006
DistilBERT FT	M	0.118	0.003	0.009	Pythia-410m FS	M	0.271	0.001	0.005
RoBERTa FT	U	0.141	0.003	0.010	Pythia-410m 0s	U	0.271	0.001	0.005
RoBERTa FT	M	0.059	0.002	0.009	Pythia-410m 0s	M	0.271	0.001	0.005
Pythia-1.4B FT	U	0.141	0.001	0.008	Pythia-160m FT	U	0.000	0.000	0.000
Pythia-1.4B FT	M	0.141	0.000	0.007	Pythia-160m FT	M	0.000	0.000	0.000
Pythia-1.4B FS	U	0.365	0.001	0.007	Pythia-160m FS	U	0.200	0.001	0.000
Pythia-1.4B FS	M	0.365	0.001	0.007	Pythia-160m FS	M	0.200	0.000	0.001
Pythia-1.4B 0s	U	0.047	0.000	0.002	Pythia-160m 0s	U	0.235	0.000	0.002
Pythia-1.4B 0s	M	0.047	0.000	0.002	Pythia-160m 0s	M	0.235	0.000	0.002
Pythia-410m FT	U	0.000	0.002	0.002					
Pythia-410m FT	M	0.000	0.002	0.002					

Composer

Model	C.	TPR	FPR	DP	Model	C.	TPR	FPR	DP
DistilBERT FT	U	0.025	0.001	0.015	Pythia-410m FS	U	0.152	0.003	0.011
DistilBERT FT	M	0.025	0.001	0.015	Pythia-410m FS	M	0.152	0.005	0.013
RoBERTa FT	U	0.003	0.000	0.013	Pythia-410m 0s	U	0.003	0.002	0.016
RoBERTa FT	M	0.083	0.002	0.013	Pythia-410m 0s	M	0.003	0.003	0.016
Pythia-1.4B FT	U	0.025	0.001	0.015	Pythia-160m FT	U	0.000	0.000	0.000
Pythia-1.4B FT	M	0.054	0.000	0.015	Pythia-160m FT	M	0.000	0.000	0.000
Pythia-1.4B FS	U	0.162	0.006	0.019	Pythia-160m FS	U	0.051	0.001	0.003
Pythia-1.4B FS	M	0.162	0.006	0.019	Pythia-160m FS	M	0.051	0.001	0.003
Pythia-1.4B 0s	U	0.067	0.003	0.011	Pythia-160m 0s	U	0.051	0.001	0.002
Pythia-1.4B 0s	M	0.067	0.003	0.011	Pythia-160m 0s	M	0.022	0.001	0.002
Pythia-410m FT	U	0.165	0.004	0.004					
Pythia-410m FT	M	0.025	0.001	0.002					

Dentist

Model	C.	TPR	FPR	DP	Model	C.	TPR	FPR	DP
DistilBERT FT	U	0.057	0.002	0.030	Pythia-410m FS	U	0.016	0.000	0.026
DistilBERT FT	M	0.057	0.002	0.030	Pythia-410m FS	M	0.016	0.001	0.027
RoBERTa FT	U	0.068	0.002	0.029	Pythia-410m 0s	U	0.006	0.002	0.026
RoBERTa FT	M	0.080	0.002	0.029	Pythia-410m 0s	M	0.006	0.002	0.025
Pythia-1.4B FT	U	0.068	0.002	0.029	Pythia-160m FT	U	0.000	0.000	0.000
Pythia-1.4B FT	M	0.068	0.002	0.029	Pythia-160m FT	M	0.000	0.000	0.000
Pythia-1.4B FS	U	0.045	0.004	0.031	Pythia-160m FS	U	0.201	0.001	0.021
Pythia-1.4B FS	M	0.045	0.004	0.030	Pythia-160m FS	M	0.198	0.001	0.024
Pythia-1.4B 0s	U	0.019	0.001	0.028	Pythia-160m 0s	U	0.016	0.000	0.016
Pythia-1.4B 0s	M	0.019	0.000	0.027	Pythia-160m 0s	M	0.027	0.001	0.016
Pythia-410m FT	U	0.082	0.002	0.019					
Pythia-410m FT	M	0.053	0.001	0.020					

Dietitian

Model	C.	TPR	FPR	DP	Model	C.	TPR	FPR	DP
DistilBERT FT	U	0.103	0.000	0.018	Pythia-410m FS	U	0.552	0.002	0.010
DistilBERT FT	M	0.103	0.000	0.018	Pythia-410m FS	M	0.552	0.002	0.010
RoBERTa FT	U	0.069	0.001	0.019	Pythia-410m 0s	U	0.586	0.001	0.013
RoBERTa FT	M	0.034	0.000	0.019	Pythia-410m 0s	M	0.552	0.001	0.012
Pythia-1.4B FT	U	0.069	0.001	0.019	Pythia-160m FT	U	0.190	0.002	0.003
Pythia-1.4B FT	M	0.069	0.000	0.018	Pythia-160m FT	M	0.017	0.011	0.013
Pythia-1.4B FS	U	0.103	0.010	0.028	Pythia-160m FS	U	0.000	0.003	0.003
Pythia-1.4B FS	M	0.103	0.010	0.028	Pythia-160m FS	M	0.000	0.004	0.004
Pythia-1.4B 0s	U	0.862	0.000	0.018	Pythia-160m 0s	U	0.000	0.001	0.001
Pythia-1.4B 0s	M	0.862	0.000	0.018	Pythia-160m 0s	M	0.000	0.001	0.001
Pythia-410m FT	U	0.034	0.006	0.007					
Pythia-410m FT	M	0.034	0.003	0.004					

Filmmaker

Model	C.	TPR	FPR	DP	Model	C.	TPR	FPR	DP
DistilBERT FT	U	0.079	0.003	0.007	Pythia-410m FS	U	0.071	0.003	0.002
DistilBERT FT	M	0.079	0.004	0.008	Pythia-410m FS	M	0.071	0.003	0.002
RoBERTa FT	U	0.064	0.003	0.008	Pythia-410m 0s	U	0.236	0.001	0.006
RoBERTa FT	M	0.150	0.002	0.006	Pythia-410m 0s	M	0.157	0.001	0.005
Pythia-1.4B FT	U	0.036	0.001	0.008	Pythia-160m FT	U	0.000	0.001	0.001
Pythia-1.4B FT	M	0.014	0.002	0.009	Pythia-160m FT	M	0.000	0.001	0.001
Pythia-1.4B FS	U	0.093	0.001	0.004	Pythia-160m FS	U	0.114	0.000	0.001
Pythia-1.4B FS	M	0.143	0.002	0.004	Pythia-160m FS	M	0.086	0.000	0.001
Pythia-1.4B 0s	U	0.071	0.000	0.001	Pythia-160m 0s	U	0.029	0.000	0.001
Pythia-1.4B 0s	M	0.071	0.000	0.001	Pythia-160m 0s	M	0.000	0.000	0.000
Pythia-410m FT	U	0.029	0.003	0.003					
Pythia-410m FT	M	0.029	0.002	0.002					

Journalist

Model	C.	TPR	FPR	DP	Model	C.	TPR	FPR	DP
DistilBERT FT	U	0.076	0.003	0.012	Pythia-410m FS	U	0.106	0.037	0.047
DistilBERT FT	M	0.064	0.006	0.009	Pythia-410m FS	M	0.092	0.025	0.035
RoBERTa FT	U	0.111	0.001	0.019	Pythia-410m 0s	U	0.025	0.018	0.025
RoBERTa FT	M	0.040	0.003	0.010	Pythia-410m 0s	M	0.047	0.021	0.027
Pythia-1.4B FT	U	0.055	0.002	0.013	Pythia-160m FT	U	0.000	0.000	0.000
Pythia-1.4B FT	M	0.033	0.007	0.006	Pythia-160m FT	M	0.000	0.000	0.000
Pythia-1.4B FS	U	0.103	0.036	0.030	Pythia-160m FS	U	0.021	0.075	0.077
Pythia-1.4B FS	M	0.090	0.034	0.028	Pythia-160m FS	M	0.007	0.085	0.086
Pythia-1.4B 0s	U	0.350	0.004	0.028	Pythia-160m 0s	U	0.137	0.002	0.011
Pythia-1.4B 0s	M	0.350	0.004	0.028	Pythia-160m 0s	M	0.074	0.002	0.008
Pythia-410m FT	U	0.009	0.002	0.003					
Pythia-410m FT	M	0.043	0.006	0.002					

Model

Model	C.	TPR	FPR	DP	Model	C.	TPR	FPR	DP
DistilBERT FT	U	0.173	0.002	0.028	Pythia-410m FS	U	0.127	0.014	0.001
DistilBERT FT	M	0.153	0.002	0.028	Pythia-410m FS	M	0.127	0.014	0.001
RoBERTa FT	U	0.153	0.002	0.023	Pythia-410m 0s	U	0.227	0.001	0.011
RoBERTa FT	M	0.173	0.003	0.030	Pythia-410m 0s	M	0.227	0.001	0.011
Pythia-1.4B FT	U	0.173	0.001	0.027	Pythia-160m FT	U	0.000	0.001	0.001
Pythia-1.4B FT	M	0.082	0.001	0.027	Pythia-160m FT	M	0.000	0.001	0.001
Pythia-1.4B FS	U	0.233	0.004	0.031	Pythia-160m FS	U	0.069	0.016	0.012
Pythia-1.4B FS	M	0.253	0.005	0.032	Pythia-160m FS	M	0.049	0.016	0.013
Pythia-1.4B 0s	U	0.067	0.004	0.009	Pythia-160m 0s	U	0.162	0.002	0.002
Pythia-1.4B 0s	M	0.067	0.004	0.009	Pythia-160m 0s	M	0.031	0.002	0.000
Pythia-410m FT	U	0.209	0.003	0.008					
Pythia-410m FT	M	0.380	0.003	0.010					

Nurse

Model	C.	TPR	FPR	DP	Model	C.	TPR	FPR	DP
DistilBERT FT	U	0.060	0.012	0.086	Pythia-410m FS	U	0.179	0.029	0.085
DistilBERT FT	M	0.129	0.014	0.076	Pythia-410m FS	M	0.235	0.028	0.085
RoBERTa FT	U	0.067	0.009	0.084	Pythia-410m 0s	U	0.056	0.020	0.084
RoBERTa FT	M	0.072	0.010	0.076	Pythia-410m 0s	M	0.034	0.021	0.083
Pythia-1.4B FT	U	0.100	0.012	0.086	Pythia-160m FT	U	0.056	0.001	0.000
Pythia-1.4B FT	M	0.034	0.009	0.079	Pythia-160m FT	M	0.033	0.001	0.000
Pythia-1.4B FS	U	0.024	0.062	0.123	Pythia-160m FS	U	0.137	0.004	0.017
Pythia-1.4B FS	M	0.024	0.062	0.123	Pythia-160m FS	M	0.107	0.006	0.015
Pythia-1.4B 0s	U	0.660	0.014	0.085	Pythia-160m 0s	U	0.260	0.006	0.030
Pythia-1.4B 0s	M	0.660	0.015	0.086	Pythia-160m 0s	M	0.260	0.007	0.031
Pythia-410m FT	U	0.113	0.005	0.040					
Pythia-410m FT	M	0.232	0.004	0.037					

Painter

Model	C.	TPR	FPR	DP	Model	C.	TPR	FPR	DP
DistilBERT FT	U	0.178	0.001	0.005	Pythia-410m FS	U	0.044	0.003	0.004
DistilBERT FT	M	0.143	0.000	0.004	Pythia-410m FS	M	0.025	0.003	0.002
RoBERTa FT	U	0.057	0.003	0.002	Pythia-410m 0s	U	0.030	0.000	0.001
RoBERTa FT	M	0.015	0.002	0.003	Pythia-410m 0s	M	0.004	0.001	0.000
Pythia-1.4B FT	U	0.043	0.000	0.002	Pythia-160m FT	U	0.000	0.001	0.001
Pythia-1.4B FT	M	0.078	0.001	0.001	Pythia-160m FT	M	0.000	0.000	0.000
Pythia-1.4B FS	U	0.001	0.000	0.000	Pythia-160m FS	U	0.018	0.001	0.001
Pythia-1.4B FS	M	0.001	0.000	0.000	Pythia-160m FS	M	0.053	0.002	0.001
Pythia-1.4B 0s	U	0.105	0.003	0.000	Pythia-160m 0s	U	0.065	0.017	0.015
Pythia-1.4B 0s	M	0.139	0.003	0.001	Pythia-160m 0s	M	0.096	0.020	0.017
Pythia-410m FT	U	0.059	0.003	0.004					
Pythia-410m FT	M	0.018	0.004	0.004					

Photographer

Model	C.	TPR	FPR	DP	Model	C.	TPR	FPR	DP
DistilBERT FT	U	0.013	0.001	0.023	Pythia-410m FS	U	0.079	0.044	0.061
DistilBERT FT	M	0.018	0.003	0.019	Pythia-410m FS	M	0.065	0.038	0.055
RoBERTa FT	U	0.054	0.002	0.029	Pythia-410m 0s	U	0.040	0.004	0.026
RoBERTa FT	M	0.089	0.002	0.026	Pythia-410m 0s	M	0.031	0.003	0.024
Pythia-1.4B FT	U	0.065	0.000	0.026	Pythia-160m FT	U	0.014	0.002	0.003
Pythia-1.4B FT	M	0.028	0.000	0.025	Pythia-160m FT	M	0.000	0.000	0.000
Pythia-1.4B FS	U	0.019	0.014	0.033	Pythia-160m FS	U	0.084	0.003	0.013
Pythia-1.4B FS	M	0.033	0.015	0.035	Pythia-160m FS	M	0.123	0.001	0.018
Pythia-1.4B 0s	U	0.057	0.008	0.008	Pythia-160m 0s	U	0.062	0.000	0.003
Pythia-1.4B 0s	M	0.057	0.008	0.008	Pythia-160m 0s	M	0.035	0.001	0.004
Pythia-410m FT	U	0.046	0.005	0.024					
Pythia-410m FT	M	0.086	0.022	0.041					

Physician

Model	C.	TPR	FPR	DP	Model	C.	TPR	FPR	DP
DistilBERT FT	U	0.087	0.006	0.027	Pythia-410m FS	U	0.055	0.033	0.020
DistilBERT FT	M	0.083	0.000	0.031	Pythia-410m FS	M	0.034	0.030	0.016
RoBERTa FT	U	0.087	0.001	0.031	Pythia-410m 0s	U	0.147	0.043	0.036
RoBERTa FT	M	0.072	0.006	0.024	Pythia-410m 0s	M	0.142	0.052	0.043
Pythia-1.4B FT	U	0.071	0.003	0.027	Pythia-160m FT	U	0.000	0.001	0.001
Pythia-1.4B FT	M	0.083	0.002	0.029	Pythia-160m FT	M	0.006	0.003	0.002
Pythia-1.4B FS	U	0.065	0.014	0.011	Pythia-160m FS	U	0.181	0.003	0.010
Pythia-1.4B FS	M	0.059	0.015	0.010	Pythia-160m FS	M	0.153	0.006	0.003
Pythia-1.4B 0s	U	0.164	0.016	0.021	Pythia-160m 0s	U	0.195	0.016	0.003
Pythia-1.4B 0s	M	0.151	0.016	0.020	Pythia-160m 0s	M	0.201	0.016	0.003
Pythia-410m FT	U	0.104	0.013	0.018					
Pythia-410m FT	M	0.129	0.015	0.018					

Poet

Model	C.	TPR	FPR	DP	Model	C.	TPR	FPR	DP
DistilBERT FT	U	0.220	0.000	0.004	Pythia-410m FS	U	0.107	0.002	0.004
DistilBERT FT	M	0.187	0.001	0.002	Pythia-410m FS	M	0.107	0.004	0.006
RoBERTa FT	U	0.153	0.001	0.003	Pythia-410m 0s	U	0.147	0.000	0.003
RoBERTa FT	M	0.227	0.001	0.005	Pythia-410m 0s	M	0.113	0.001	0.001
Pythia-1.4B FT	U	0.120	0.001	0.003	Pythia-160m FT	U	0.000	0.000	0.000
Pythia-1.4B FT	M	0.187	0.002	0.005	Pythia-160m FT	M	0.000	0.000	0.000
Pythia-1.4B FS	U	0.153	0.001	0.003	Pythia-160m FS	U	0.060	0.003	0.004
Pythia-1.4B FS	M	0.153	0.000	0.003	Pythia-160m FS	M	0.060	0.004	0.005
Pythia-1.4B 0s	U	0.133	0.002	0.004	Pythia-160m 0s	U	0.020	0.008	0.009
Pythia-1.4B 0s	M	0.133	0.003	0.005	Pythia-160m 0s	M	0.093	0.008	0.010
Pythia-410m FT	U	0.060	0.001	0.002					
Pythia-410m FT	M	0.127	0.004	0.002					

Professor

Model	C.	TPR	FPR	DP	Model	C.	TPR	FPR	DP
DistilBERT FT	U	0.024	0.012	0.022	Pythia-410m FS	U	0.040	0.006	0.018
DistilBERT FT	M	0.015	0.011	0.024	Pythia-410m FS	M	0.042	0.005	0.018
RoBERTa FT	U	0.018	0.024	0.032	Pythia-410m 0s	U	0.299	0.029	0.118
RoBERTa FT	M	0.012	0.024	0.034	Pythia-410m 0s	M	0.301	0.028	0.118
Pythia-1.4B FT	U	0.003	0.012	0.030	Pythia-160m FT	U	0.046	0.011	0.005
Pythia-1.4B FT	M	0.003	0.002	0.023	Pythia-160m FT	M	0.006	0.017	0.014
Pythia-1.4B FS	U	0.120	0.010	0.050	Pythia-160m FS	U	0.097	0.019	0.047
Pythia-1.4B FS	M	0.119	0.010	0.050	Pythia-160m FS	M	0.120	0.024	0.057
Pythia-1.4B 0s	U	0.078	0.243	0.206	Pythia-160m 0s	U	0.144	0.231	0.211
Pythia-1.4B 0s	M	0.083	0.242	0.207	Pythia-160m 0s	M	0.165	0.215	0.206
Pythia-410m FT	U	0.020	0.026	0.007					
Pythia-410m FT	M	0.020	0.026	0.008					

Psychologist

Model	C.	TPR	FPR	DP	Model	C.	TPR	FPR	DP
DistilBERT FT	U	0.006	0.002	0.023	Pythia-410m FS	U	0.060	0.001	0.010
DistilBERT FT	M	0.010	0.003	0.022	Pythia-410m FS	M	0.067	0.001	0.010
RoBERTa FT	U	0.034	0.000	0.024	Pythia-410m 0s	U	0.024	0.004	0.014
RoBERTa FT	M	0.058	0.003	0.027	Pythia-410m 0s	M	0.013	0.004	0.015
Pythia-1.4B FT	U	0.060	0.002	0.025	Pythia-160m FT	U	0.011	0.002	0.002
Pythia-1.4B FT	M	0.031	0.002	0.024	Pythia-160m FT	M	0.000	0.001	0.001
Pythia-1.4B FS	U	0.010	0.023	0.038	Pythia-160m FS	U	0.026	0.005	0.005
Pythia-1.4B FS	M	0.010	0.023	0.038	Pythia-160m FS	M	0.008	0.003	0.004
Pythia-1.4B 0s	U	0.305	0.006	0.030	Pythia-160m 0s	U	0.003	0.000	0.002
Pythia-1.4B 0s	M	0.294	0.006	0.029	Pythia-160m 0s	M	0.028	0.000	0.002
Pythia-410m FT	U	0.175	0.011	0.025					
Pythia-410m FT	M	0.120	0.006	0.017					

Software Engineer

Model	C.	TPR	FPR	DP	Model	C.	TPR	FPR	DP
DistilBERT FT	U	0.070	0.010	0.028	Pythia-410m FS	U	0.021	0.003	0.003
DistilBERT FT	M	0.112	0.010	0.027	Pythia-410m FS	M	0.043	0.003	0.004
RoBERTa FT	U	0.112	0.009	0.026	Pythia-410m 0s	U	0.155	0.006	0.013
RoBERTa FT	M	0.052	0.009	0.028	Pythia-410m 0s	M	0.112	0.006	0.012
Pythia-1.4B FT	U	0.027	0.009	0.029	Pythia-160m FT	U	0.000	0.001	0.001
Pythia-1.4B FT	M	0.036	0.007	0.029	Pythia-160m FT	M	0.000	0.001	0.001
Pythia-1.4B FS	U	0.322	0.040	0.061	Pythia-160m FS	U	0.021	0.002	0.003
Pythia-1.4B FS	M	0.322	0.040	0.061	Pythia-160m FS	M	0.021	0.004	0.005
Pythia-1.4B 0s	U	0.103	0.014	0.027	Pythia-160m 0s	U	0.000	0.001	0.001
Pythia-1.4B 0s	M	0.103	0.014	0.027	Pythia-160m 0s	M	0.000	0.001	0.001
Pythia-410m FT	U	0.106	0.001	0.004					
Pythia-410m FT	M	0.043	0.000	0.001					

Surgeon

Model	C.	TPR	FPR	DP	Model	C.	TPR	FPR	DP
DistilBERT FT	U	0.070	0.010	0.041	Pythia-410m FS	U	0.045	0.002	0.010
DistilBERT FT	M	0.059	0.008	0.038	Pythia-410m FS	M	0.007	0.002	0.010
RoBERTa FT	U	0.043	0.011	0.042	Pythia-410m 0s	U	0.018	0.001	0.010
RoBERTa FT	M	0.014	0.006	0.034	Pythia-410m 0s	M	0.018	0.002	0.011
Pythia-1.4B FT	U	0.116	0.012	0.045	Pythia-160m FT	U	0.000	0.000	0.000
Pythia-1.4B FT	M	0.055	0.009	0.041	Pythia-160m FT	M	0.000	0.000	0.000
Pythia-1.4B FS	U	0.002	0.013	0.030	Pythia-160m FS	U	0.016	0.001	0.002
Pythia-1.4B FS	M	0.002	0.013	0.030	Pythia-160m FS	M	0.004	0.001	0.002
Pythia-1.4B 0s	U	0.116	0.001	0.000	Pythia-160m 0s	U	0.000	0.000	0.000
Pythia-1.4B 0s	M	0.104	0.001	0.001	Pythia-160m 0s	M	0.000	0.000	0.000
Pythia-410m FT	U	0.198	0.011	0.030					
Pythia-410m FT	M	0.252	0.005	0.024					

Teacher

Model	C.	TPR	FPR	DP	Model	C.	TPR	FPR	DP
DistilBERT FT	U	0.073	0.000	0.021	Pythia-410m FS	U	0.157	0.033	0.045
DistilBERT FT	M	0.028	0.006	0.024	Pythia-410m FS	M	0.170	0.035	0.047
RoBERTa FT	U	0.132	0.010	0.033	Pythia-410m 0s	U	0.227	0.202	0.220
RoBERTa FT	M	0.133	0.005	0.028	Pythia-410m 0s	M	0.227	0.208	0.225
Pythia-1.4B FT	U	0.099	0.010	0.032	Pythia-160m FT	U	0.000	0.000	0.000
Pythia-1.4B FT	M	0.178	0.009	0.034	Pythia-160m FT	M	0.000	0.000	0.000
Pythia-1.4B FS	U	0.228	0.054	0.074	Pythia-160m FS	U	0.037	0.004	0.008
Pythia-1.4B FS	M	0.228	0.054	0.074	Pythia-160m FS	M	0.056	0.007	0.012
Pythia-1.4B 0s	U	0.534	0.060	0.091	Pythia-160m 0s	U	0.447	0.205	0.225
Pythia-1.4B 0s	M	0.534	0.061	0.092	Pythia-160m 0s	M	0.513	0.205	0.228
Pythia-410m FT	U	0.043	0.001	0.005					
Pythia-410m FT	M	0.053	0.010	0.017					

D Gender-Conditioned Faithfulness by Profession

Mean faithfulness scores split by gender for key professions. Only professions with at least 10 examples per gender group are shown. DistilBERT unmasked condition; n = number of examples.

Profession	Female		Male	
	Faith.	n	Faith.	n
Nurse	0.354	132	**0.756**	15
Dietitian	0.656	27	0.288	3
Psychologist	0.370	79	0.336	55
Physician	0.198	171	0.270	157
Attorney	0.161	110	0.162	160
Professor	0.158	413	0.162	519
RoBERTa unmasked condition				
Nurse	0.163	130	**0.534**	16
Dietitian	0.549	28	0.533	2
Psychologist	0.234	77	0.227	52
Physician	0.280	174	0.184	154
Attorney	0.192	108	0.164	156
Professor	0.155	411	0.168	533

The nurse|M asymmetry is the largest gender-conditioned faithfulness gap observed: DistilBERT 0.756 vs. 0.354, RoBERTa 0.534 vs. 0.163. In both models, male nurse predictions depend on a critically small set of attributed tokens.

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